

# QUESTION 1 – SINGLE INHERITANCE

## Java Inheritance Experiment

### 1. Objective

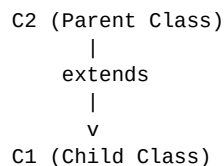
To implement Single Inheritance in Java, demonstrate how a child class inherits methods from a parent class, execute the program, and verify the result using multiple test cases.

### 2. Task Description

Implement the given inheritance program with suitable explanations as code comments. Execute the program, verify the output with multiple test cases, and include the source code, explanation, UML diagram, and output verification in the PDF.

### 3. Inheritance Type

**Single Inheritance:** A single child class inherits from one parent class. In this program, C1 extends C2.



### 4. Step-wise Execution

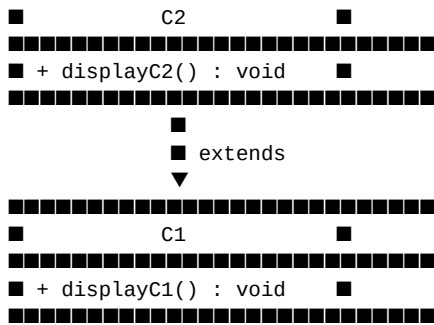
1. Define class C2 as the parent class.
2. Create displayC2() inside C2 to display the parent-class message.
3. Define class C1 using extends C2, making C1 the child class.
4. Create displayC1() inside C1.
5. Create an object of C1 in the main() method.
6. Use the C1 object to call displayC2(), which is inherited from C2.
7. Use the same object to call displayC1().
8. Compare the actual output with the expected output.

### 5. Algorithm

1. Start the program.
2. Create the parent class C2 and define displayC2().
3. Create child class C1 by extending C2 and define displayC1().
4. Create an object obj of class C1.
5. Call displayC2() using obj.
6. Call displayC1() using obj.
7. Display the results.
8. Stop the program.

### 6. UML Diagram





## 7. Java Program

```

// Parent class
class C2 {

    // Method of parent class
    void displayC2() {
        System.out.println("This is Class C2");
    }
}

// Child class C1 inherits the methods of C2.
// This represents Single Inheritance because
// one child class inherits from one parent class.
class C1 extends C2 {

    // Method of child class
    void displayC1() {
        System.out.println("This is Class C1");
    }
}

// Main class
public class Question1 {

    public static void main(String[] args) {

        // Creating an object of child class C1
        C1 obj = new C1();

        // Calling the inherited method from parent class C2
        obj.displayC2();

        // Calling the method of child class C1
        obj.displayC1();
    }
}
  
```

## 8. Output

```

This is Class C2
This is Class C1
  
```

**Output verification:** The child-class object successfully accesses both its own method and the inherited parent-class method.

## 9. Multiple Test Cases

| Test Case | Operation                  | Expected Output        | Status |
|-----------|----------------------------|------------------------|--------|
| 1         | displayC2() → displayC1()  | C2 message, C1 message | PASS   |
| 2         | displayC1() → displayC2()  | C1 message, C2 message | PASS   |
| 3         | Call inherited displayC2() | C2 message             | PASS   |

## 10. Test Case Details

**Test Case 1:** Calling displayC2() followed by displayC1() produces the expected C2 and C1 messages. Result: PASS.

**Test Case 2:** Reversing the method-call order produces the expected C1 and C2 messages. Result: PASS.

**Test Case 3:** Calling displayC2() through a C1 object confirms that the parent method is inherited. Result: PASS.

## 11. Result

The Java program was successfully implemented and demonstrates Single Inheritance. Class C1 inherits the displayC2() method from class C2, and all listed test cases produce the expected results.

## 12. Conclusion

Single Inheritance allows a child class to reuse the properties and methods of one parent class. The program successfully demonstrates this concept using classes C2 and C1.